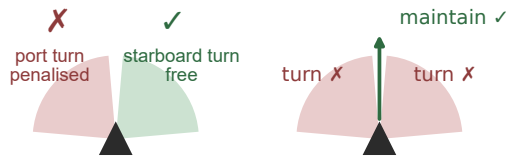


(a)

Directional non-compliance penalty

Head-On · Give-Way · Overtaking

Stand-On

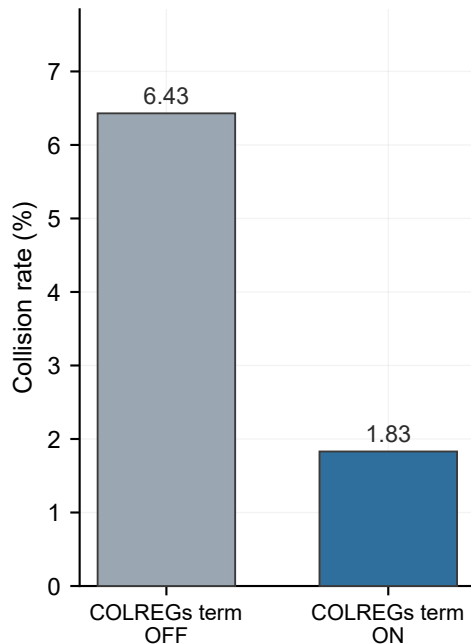


$$R_i^{\text{COLREGs}} = -\alpha_c (1 + \rho_i) \mathbf{1}[\rho_i > 0.3] (\dots)$$

A directional penalty on non-compliant turning — not a positive compliance bonus.
Weighted by collision risk ρ_i and applied only above a risk gate.

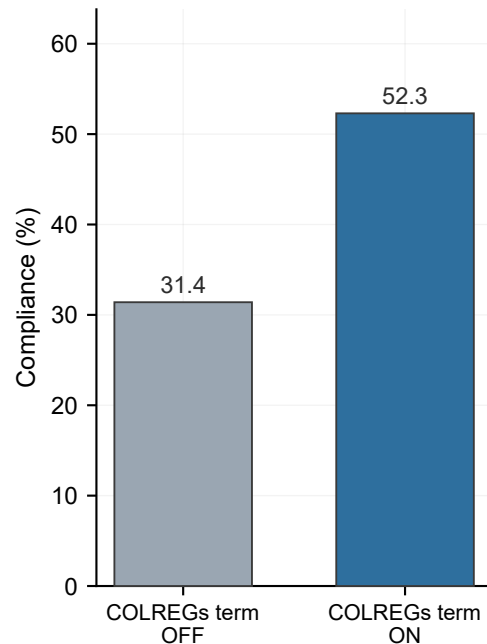
(b)

Collision rate



(c)

Overall COLREGs compliance



Matched proposed-architecture comparison, three seeds per arm (OFF: qo_SE_COLREGSOFF_s42/43/44, ON: qd_MOE_SE_s42/43/44). Per-seed dots require the source logs.
The COLREGs term acts as a domain prior that breaks symmetric but mutually incompatible avoidance choices, improving both rule compliance and collision safety.